

Prior Probabilities 1

Prior probabilities are the initial, unconditional probabilities assigned to an event or parameter before new evidence or data is taken into account.

It represents existing knowledge or beliefs, which are updated using Bayesian inference to form posterior probabilities.

Priors are crucial in decision-making, AI, and statistics to frame initial expectations.

$$P(H|e) = \frac{P(e|H)P(H)}{P(e)}$$

$P(H)$ is the prior probabilities,

$P(e|H)$ is the likelihood

$P(H|e)$ is the posterior probabilities

